# Chapter 14 Notes

## Section 1

### Notes

Let $K$ and $K'$ be classes, $K$ be transitive, $K \subseteq K'$, and $\phi(x_1, ..., x_n)$ a formula whose constants are all in $K$. Note that, even if we assume $\phi(a_1, ..., a_n)$ is true over $K$ iff it is true over $K'$ (for all constants $a_1, ..., a_n$ in $K$), we *cannot* prove that $(\exists x)(\phi(x, x_1, ..., x_n))$ is such that $(\exists x)(\phi(x, a_1, ..., a_n))$ is true over $K$ iff it is true over $K'$ (for all constants $a_1, ..., a_n$ in $K$); call any $\phi$ *good* if this property does hold for it. For, let $a_1, ..., a_n$ be constants in $K$. On the one hand, suppose that $(\exists x)(\phi(x, a_1, ..., a_n))$ is true over $K$. Then for some $a \in K$, $\phi(a, a_1, ..., a_n)$ is true over $K$, and thus by assumption, $\phi(a, a_1, ..., a_n)$ is true over $K'$, hence $(\exists x)(\phi(x, a_1, ..., a_n))$ is true over $K'$. However, on the other hand, suppose that $(\exists x)(\phi(x, a_1, ..., a_n))$ is true over $K'$. Then for some $a \in K'$, $\phi(a, a_1, ..., a_n)$ is true over $K'$. However, unlike in Exercise 1.1 (a), we *cannot* conclude that $a \in K$, meaning it does not necessarily follow that $\phi(a, a_1, ..., a_n)$ is true over $K$ (and thus it does not necessarily follow that $(\exists x)(\phi(x, a_1, ..., a_n))$ is true over $K$.

Using the same definitions as in the preceeding paragraph, note the following. Suppose $\phi(x_1, ..., x_n)$ is only such that if $\phi(a_1, ..., a_n)$ is true over $K$ (for constants $a_1, ..., a_n$ in $K$), then $\phi(a_1, ..., a_n)$ is true over $K'$. It certainly does *not* follow that $\sim\phi$ is good (or even satisfies the weaker property if if $\sim\phi(a_1, ..., a_n)$ is true over $K$, then it is true over $K'$). For, if $\sim\phi(a_1, ..., a_n)$ is true over $K$, then $\phi(a_1, ..., a_n)$ is false over $K$. But because we do not have that $\phi$ is good, we cannot conclude that $\phi(a_1, ..., a_n)$ is false over $K'$ (now, if $\sim\phi(a_1, ..., a_n)$ was true over $K'$, then $\phi(a_1, ..., a_n)$ would be false over $K'$, and then by assumption, $\phi(a_1, ..., a_n)$ would also need to be false over $K$, hence $\sim\phi(a_1, ..., a_n)$ would be true over $K$).

The above paragraph explains why $\Sigma$-formulas do not include negations (unless the subformlula the negation is the main operator of is itself a $\Sigma_0$ formula).

### Proposition 1.1

Note that part (3) of this proposition follows in a similar way as part (3) of Exercise 1.1 (b) is proven below.

### Exercise 1.1 (a)

Suppose $K$ and $K'$ are classes, such that $K$ is transitive, $K \subseteq K'$, $a_1, ..., a_n$ are elements of $K$, and $\phi(x_1, ..., x_n)$ is a formula whose constants are all in $K$. Suppose also that $\phi(x_1, ..., x_n)$ is $\Sigma_0$. We are to show that $\phi(a_1, ..., a_n)$ is true over $K$ iff it is true over $K'$. We prove this by induction on the degree of $\phi(x_1, ..., x_n)$.

Base Case: If $\phi(x_1, ..., x_n)$ is of the form $x \in y$, then $a_1 \in a_2$ is true over $K$ iff it is true over $K'$ (as $a_1$ and $a_2$ are both in $K'$, as $K \subseteq K'$).

If the result holds for some $\phi$, then obviously it also holds for $\sim\phi$.

If the result holds for $\phi$ and $\psi$, then obviously it also holds for $\phi \wedge \psi$.

Suppose the result holds for some $\phi(x, y, x_1, ..., x_n)$ (whose free variables are all in the set $\{x, y, x_1, ..., x_n\}$). We must show that the result also holds for

$$(\exists x \in y)(\phi(x, y, x_1, ..., x_n))$$

Take any constants $a, b_1, ..., b_n$ of $K$. We must show that $(\exists x \in a)(\phi(x, a, b_1, ..., b_n))$ is true over $K$ iff it is true over $K'$. On the one hand, suppose that it is true over $K$. Then for some $b \in a$, $\phi(b, a, b_1, ..., b_n)$ is true over $K$. But the constants $b, a, b_1, ..., b_n$ are all in $K$, meaning by assumption, $\phi(b, a, b_1, ..., b_n)$ is true over $K'$, and thus $(\exists x \in a)(\phi(x, a, b_1, ..., b_n))$ is true over $K'$. On the other hand, suppose that $(\exists x \in a)(\phi(x, a, b_1, ..., b_n))$ is true over $K'$. Then for some $b \in a$, $\phi(b, a, b_1, ..., b_n)$ is true over $K'$. But $K$ is transitive, and $a \in K$, meaning $b \in K$. Thus the constants $b, a, b_1, ..., b_n$ are all in $K$, meaning, by assumption, $\phi(b, a, b_1, ..., b_n)$ is true over $K$, and thus $(\exists x \in a)(\phi(x, a, b_1, ..., b_n))$ is true over $K$.

### Exercise 1.1 (b)

Suppose $K$ and $K'$ are classes, such that $K$ is transitive, $K \subseteq K'$, $a_1, ..., a_n$ are elements of $K$, and $\phi(x_1, ..., x_n)$ is a formula whose constants are all in $K$. Suppose also that $\phi(x_1, ..., x_n)$ is $\Sigma$. We are to show that $\phi(a_1, ..., a_n)$ is true over $K$ iff it is true over $K'$. Call a formula $\phi$ good if it satisfies this result. We prove this by induction on the degree of $\phi(x_1, ..., x_n)$. In fact, we prove:

(1) Every $\Sigma_0$ formula is good.

(2) If $\phi, \psi$ are good, then so are $\phi \wedge \psi$ and $\phi \vee \psi$.

(3) If $\phi$ is good, so is $(\exists x)\phi$.

(4) If $\phi$ is good, so are $(\forall x \in y)\phi$ and $(\exists x \in y)\phi$.

*Proof.* (1) This follows by Exercise 1.1 (a).

(2) Obvious.

(3) Suppose $\phi(x, x_1, ..., x_n)$ is good (where all free variables of $\phi$ are in the set $\{x, x_1, ..., x_n\}$). We must show that $(\exists x)\phi(x, x_1, ..., x_n)$ is good. So suppose $(\exists x)\phi(x, a_1, ..., a_n)$ was true over $K$. Then, for some $a \in K$, $\phi(a, a_1, ..., a_n)$ is true over $K$, hence by assumption, $\phi(a, a_1, ..., a_n)$ is true over $K'$, meaning $(\exists x)\phi(x, a_1, ..., a_n)$ is true over $K'$.

(4) Suppose $\phi(x, y, z_1, ..., z_n)$ is good. We wish to show that $(\forall x \in y)\phi(x, y, z_1, ..., z_n)$ is also good. Well, let $a, b_1, ..., b_n$ be constants in $K$. Suppose $(\forall x \in a)\phi(x, a, b_1, ..., b_n)$ is true over $K$. Because $K$ is transitive, it follows that $\phi(c, a, b_1, ..., b_n)$ is true over $K$ for all elements $c \in a$. Then, by the assumption that $\phi$ is good, it follows that $(\forall x \in a)\phi(x, a, b_1, ..., b_n)$ is true over $K'$.

As for the formula $(\exists x \in y)\phi$, this is $(\exists x)(x \in y \wedge \phi)$, which is good by parts (1), (2), and (3) of this exercise. □

## Section 2

### Notes

It is remarked in the last paragraph of this section that once we show $y = \mathcal{F}(x)$ is $\Sigma$, it will follow by Theorem 2.8 that the relation $y = L_\alpha$ is also $\Sigma$, and absolute. It does follow that it is $\Sigma$, but it does not immediately follow by the theorem that it is absolute - only that it is *upwards* absolute.

### Propositions 2.6 and 2.7

There are multiple problems with these two propositions. First, the propositions, as stated, are clearly incorrect, as the proofs given require $f$ to be $\Sigma$, which is not assumed (because $f$ here is being treated as a constant, not a variable). To fix this, for any set $x$, let us define $x''(y)$ for any set $y$ to be the set of all $z \in \text{Ran}(x)$ such that $\langle w, z \rangle \in x$ for some $w \in y$. This more general definiton allows us to correct the proof.

In particular, we show that the relation $y = z''(x)$ (this time, $z$ is a *variable*, meaning this is a legitimate equation, without any prior assumptions of being $\Sigma$) is $\Sigma$ (in fact $\Sigma_0$).

*Proof.* $y = z''(x)$ is the conjunction of the following two $\Sigma_0$ conditions:

$$(1)\ (\forall v \in x)(\forall w \in z)(\langle v, w\rangle \in z \supset w \in y)$$
$$(2)\ (\forall v \in y)(\exists w \in x)(\langle w, v\rangle \in z)$$

□

If we were to restrict $z$ to functions, then the proof would involve adding the condition that $z$ is a function (that is, add in $\text{Fun}(z)$, which is $\Sigma_0$ by results from Chapter 12).

### Theorem 2.8

There is an issue with the proof of this theorem. In particular, $f(\beta + 1) = F(f \restriction \beta)$ is used in the supposed $\Sigma$ formula, yet it was never established that $f(\beta + 1) = F(f \restriction \beta)$ is itself $\Sigma$. Here is why this is.

First, note that the relation $y = f \restriction x$ is $\Sigma$, as it is equivalent to the conjunction of the following two formulas:

$$(1)\ (\forall z \in y)(\exists w \in \text{Dom}(f))(\exists v \in \text{Ran}(f))(w \in x \wedge z = \langle w, v\rangle)$$
$$(2)\ (\forall z \in f)((\exists w \in x)(\exists v \in \text{Ran}(f))(z = \langle w, v\rangle) \supset z \in y)$$

It is obvious that the above result means $f(\beta + 1) = F(f \restriction \beta)$ is $\Sigma$. To be specific, it is then represented by the following $\Sigma$ formula:

$$(1)\ (\exists z \in \text{Ran}(f))(\exists y)(\exists w)(\langle \beta + 1, z\rangle \in f \wedge w = f \restriction \beta \wedge y = F(w) \wedge y = z)$$

There is another error in the proof of this theorem. The occurences of $(\forall \beta \in \alpha + 1)$ should be replaced by $(\forall \beta \in \alpha)$ for the following reason: If we took the previous formula $((\forall \beta \in \alpha + 1))$, the formula in the proof would tell us that, for $\alpha$, $f(\alpha + 1) = F(f \restriction \alpha)$, meaning $f(\alpha + 1)$ is defined. But that is not how we defined $f$, as we needed $\text{Dom}(f) = \alpha + 1$, meaning $\alpha + 1$ would *not* be in the domain of $f$.

In fact, it seems that $F(f \restriction \beta)$ is just a typo, and really should be $F(f(b))$. This is due to the definition of $F^*$: $F^*(\alpha + 1)$ is not equal to $F(F^* \restriction \alpha)$, but is equal to $F(F^*(\alpha))$.

## Section 3

### Proposition 3.3

Parts (3) and (4) of this proposition need some clarification. In particular, they are using $n$ as a constant, not a variable. This works, as we recall that it was shown in Chapter 12 that $x = n, n \in x, x \in n$ are all $\Sigma_0$ (hence $\Sigma$).

### Types

It is remakred that the conditions of (0) can be collectively stated as $t(m_1 \in m_2) = \{\lceil m_1 \rceil, \lceil m_2 \rceil\} \cap \omega$. However, this is not correct. For instance, $t(x_i \in x_j) = \{i, j\}$, but according to the other definition, it would equal

$$\{\lceil m_1 \rceil, \lceil m_2 \rceil\} \cap \omega$$

$$= \{\langle 0, i\rangle, \langle 0, j\rangle\} \cap \omega$$

which may not be equal to $\{i, j\}$.

## Lemma 3.4

As remarked in the above Types section, the formula (2) is not correct. One way to correct this is by replacing it by the following method. First, we need a relation $\mathrm{Var}(x)$ (meaning $x$ is the code of some variable). This is just:

$$(\exists y \in x)(\exists z \in y)(\mathrm{Num}(z) \wedge x = \langle 0, z\rangle)$$

which is $\Sigma_0$. Then we need the relation $\mathrm{Var}^n(x)$, meaning $x$ is the code of the variable $x_n$. This is just:

$$(\exists y \in x)(\exists z \in y))(z = n \wedge x = \langle 0, z\rangle)$$

which is $\Sigma_0$. Then we change (2) in the proof of the proposition to the conjunction of the following four formulas (for the sake of space, let "$\epsilon$" stand for $(\forall x \in c(a))(\forall y \in c(a))$):

(1) $\epsilon((\mathrm{Var}(x) \wedge \mathrm{Var}(y)) \supset$
$(\exists i \in \omega)(\exists j \in \omega)(\mathrm{Var}^i(x) \wedge \mathrm{Var}^j(y) \wedge f(\langle x, y, 0\rangle) = \{i, j\}))$
(2) $\epsilon((\mathrm{Var}(x) \wedge \sim\mathrm{Var}(y)) \supset$
$(\exists i \in \omega)(\mathrm{Var}^i(x) \wedge f(\langle x, y, 0\rangle) = \{i\rangle\}))$
(3) $\epsilon((\sim\mathrm{Var}(x) \wedge \mathrm{Var}(y)) \supset$
$(\exists i \in \omega)(\mathrm{Var}^i(y) \wedge f(\langle x, y, 0\rangle) = \{i\rangle\}))$
(4) $\epsilon((\sim\mathrm{Var}(x) \wedge \sim\mathrm{Var}(y)) \supset f(\langle x, y, 0\rangle) = \emptyset)$

We can use $\sim\mathrm{Var}(x)$ in the above formulas, as $\mathrm{Var}(x)$ is $\Sigma_0$, meaning so is $\sim\mathrm{Var}(x)$.